# Odd-Rank Loci and Box-Induced Symmetries of Centered Sudoku Operators

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**Abstract**

We study how the rank of a Latin square changes when its symbols are relabelled. To a square $L$ and a relabeling $\gamma \in S_n$ we attach the centered matrix $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2} J_n$ and restrict it to the standard hyperplane. For Sudoku grids, the box constraint becomes an exact linear identity: averaging $E_{\gamma,L}$ over row-bands and column-stacks gives zero for every relabeling, while a cyclic Latin-square witness shows that the identity is not forced by Latin-squareness alone. Kernel lines on the standard hyperplane admit primitive integral representatives in the root lattice $A_{n-1}$; the box identities select structured such certificates in the order-9 examples.

The paper has two kinds of results. The analytic part proves the Box Band Lemma, the kernel-lift lemma, and a general stabilizer/lattice dichotomy for two sparse root-lattice templates at every $n \geq 5$. The certified finite part exhaustively studies selected orders and laboratory bases. In particular, for a Sudoku grid $M_{19}$ of order 9, a distinguished 72-element subset of the rank-drop locus is exactly the set of relabelings whose middle row-band has column sums 15. This same subset is a single left and right coset, with affine stabilizers isomorphic to $3^2{:}D_4$, and it is also the isomorphism torsor from the reduced middle-band hypergraph of $M_{19}$ to a six-edge subhypergraph of the magic triples. The underlying Latin square has trivial autotopism group, so this affine symmetry is not a classical Latin-square symmetry. All finite claims are accompanied by a reproducibility package with exact-arithmetic certificates and a SHA-256 manifest.

**Keywords.** Sudoku, Latin squares, centered operators, odd-rank locus, $A_{n-1}$ root lattice, affine group, autotopism.

**MSC 2020.** 05B15 (primary), 15A03, 20B25, 11C20, 05E18.

## 1 Introduction

### 1.1 From Latin squares to Sudoku via centered operators

Let $L$ be a Latin square of order $n$ on the symbol set $\{1, \dots, n\}$. The *centered operator* associated with a symbol relabeling $\gamma \in S_n$ is

$$E_{\gamma,L} \;:=\; \gamma(L) \;-\; \frac{n+1}{2}\, J_n, \tag{1}$$

where $\gamma(L)$ is the entrywise application of $\gamma$ and $J_n$ is the $n \times n$ all-ones matrix. The operator $E_{\gamma,L}$ has constant row and column sums equal to zero, hence restricts to an endomorphism of the standard hyperplane $V_{\mathrm{std},n} = \{x \in \mathbb{Q}^n : \mathbf{1}^\top x = 0\}$.

Three observations motivate the present paper. First, the centering in (1) is the same as in the sibling determinant-divisibility paper [1]; this paper studies the *rank* of $E_{\gamma,L}$, not its determinant. Second, the rank is an integer-valued invariant of the pair $(L, \gamma)$ which is generically equal to $n-1$ on $V_{\mathrm{std},n}$ but can drop to $n-2$ on a sparse, structured subset of $S_n$. Third, when $L$ is *Sudoku* (i.e. also satisfies the $\sqrt{n} \times \sqrt{n}$ box constraint at $n = 9$), the rank-drop locus and its kernel certificates exhibit additional algebraic structure that is not forced by the Latin-square axioms alone.

### 1.2 Why box constraints matter

The box constraint is usually a combinatorial requirement: every $\sqrt{n} \times \sqrt{n}$ block of a Sudoku grid contains the symbols $\{1, \dots, n\}$. We promote it to an exact linear identity. Let $B_{\mathrm{rb}}, B_{\mathrm{cs}} \in \{0,1\}^{n \times \sqrt{n}}$ be the row-band and column-stack incidence matrices, and let $P_{\mathrm{rb}} := \frac{1}{\sqrt{n}} B_{\mathrm{rb}}^{\top}$, $P_{\mathrm{cs}} := \frac{1}{\sqrt{n}} B_{\mathrm{cs}}^{\top} \in \mathbb{Q}^{\sqrt{n} \times n}$ be the corresponding band-mean and stack-mean averaging maps (defined precisely in §3.1). Then for every Sudoku grid $L$ and every $\gamma \in S_n$,

$$P_{\mathrm{rb}}\, E_{\gamma,L}\, P_{\mathrm{cs}}^{\top} \;=\; 0. \tag{2}$$

This identity is not a consequence of Latin-squareness alone; the explicit cyclic witness in §3.3 isolates the box contribution at the linear-algebraic level. The full formal statement of (2) is Theorem 3.1 below.

### 1.3 Main results

The four main contributions of the paper are stated informally below; each appears in formal form exactly once in the dedicated section indicated.

**(a)** **Box Band Lemma (§3).** The identity $P_{\mathrm{rb}} E_{\gamma,L} P_{\mathrm{cs}}^{\top} = 0$ is an exact, Sudoku-specific linear identity. It is not forced by the Latin-square axioms alone. *(Theorem 3.1, certified.)*

**(b)** **Odd-rank certificates lie in $A_{n-1}$ (§6, §7).** For every relabeling $\gamma$ producing odd rank $n-2$ on $V_{\mathrm{std},n}$, the rational kernel line lifts to $V_{\mathrm{std},n}$ and has a primitive representative in the root lattice $A_{n-1} = V_{\mathrm{std},n} \cap \mathbb{Z}^n$. This is the natural centered-operator certificate framework; at $n = 9$, the distinguished sparse height-one and height-two template orbits $M$ and $O_6$ saturate $A_8$: $\Lambda_M = \Lambda_{O_6} = A_8$. *(Theorem 6.1; Certified Proposition 7.1.)*

**(c)** **Affine coset symmetry of $M_{19}$ (§8).** On the Sudoku base $M_{19}$, the left/right set-stabilizers of the shared left-kernel subset $\Gamma^*_{M_{19}}$ satisfy $K_L \cong K_R \cong 3^2{:}D_4$, and the subset $\Gamma^*_{M_{19}}$ (of size 72, sitting inside the full rank-locus $\Gamma_{\mathrm{rank}}(M_{19})$ of size 100) form a single $K_L$-coset $\Gamma^*_{M_{19}} = K_L \pi_0 = \pi_0 K_R$ with $K_L = \pi_0 K_R \pi_0^{-1}$. Equivalently, $\Gamma^*_{M_{19}}$ is exactly the set of middle-band balanced relabelings of $M_{19}$: in every column, the entries in rows $4, 5, 6$ sum to 15 after applying $\gamma$. The number $4320 = 3! \cdot 6!$ is the order of the ambient coordinate stabilizer $\mathrm{Stab}_{S_9}(w_9^L) \cong S_3 \times S_6$, not the size of $\Gamma^*_{M_{19}}$. *(Proposition 10.4; Certified Proposition 8.5; Certified Proposition 8.4.)*

**(d)** **Sudoku-specificity beyond autotopisms (§10).** The Latin square underlying $M_{19}$ is rigid as a Latin square: $|\,\mathrm{Aut}(L_{M_{19}})| = 1$. Therefore $K_L$ is *not* the image of any autotopism subgroup. In this grid, the affine coset structure is detected by the Sudoku middle-band balance condition. *(Certified Proposition 10.1; Corollary 10.2.)*

We complement *(a)*–*(d)* with an exhaustive computational verification of a stabilizer dichotomy for the odd-rank locus representatives $\{M_n, O_n\}$ at $5 \le n \le 13$ (Proposition 9.1), plus an analytic certified identification of the abstract isomorphism types $\mathrm{Stab}(M_n) \cong D_4 \times S_{n-4}$ and $\mathrm{Stab}(O_n) \cong C_2 \times S_{n-4}$ at $n \in \{7, 9, 11\}$ (Certified Proposition 9.2). The same stabilizer and lattice-saturation dichotomy is proved for every $n \ge 5$ in Theorem 9.3.

**Terminology for finite certificates.** We reserve the word *Theorem* for statements proved in the text by ordinary mathematical argument. Statements whose proof is an exhaustive finite computation, with exact arithmetic and a reproducible certificate, are labelled *Certified Proposition*. This distinction is editorial, not epistemic: the certified propositions are part of the paper's claimed results, but their proof object is the checked certificate package rather than a hand derivation alone. Section 11 describes the verification protocol.

### 1.4 Why this structure matters

The main point is not that a few relabelings happen to lower rank. It is that the Sudoku box constraint produces exact linear relations after centering, and those relations interact with the natural root-lattice certificate framework for centered operators. This gives a way to separate ordinary Latin-square phenomena from box-induced phenomena: Latin squares need not satisfy the band/stack identity, while Sudoku grids do so uniformly in the symbol relabeling.

The $M_{19}$ computation is included because it exposes a second layer of structure. A rank-drop subset that first appears as an exhaustive list is recovered as a middle-band balance condition, then as a single affine coset with stabilizer $3^2{:}D_4$, and finally as a hypergraph isomorphism torsor. Since the underlying Latin square has trivial autotopism group, this symmetry is not explained by the classical row, column, and symbol symmetries of Latin squares. The certified finite results therefore serve as a controlled laboratory for the general lattice mechanisms proved later in the paper.

### 1.5 Relation to prior work

The classical Sudoku and Latin-square literature works along axes distinct from ours: completion complexity [4], enumeration [10, 15], minimum-clue problems [13], group-based and orthogonal constructions [7, 17], autotopism computation [16, 6], and Latin-square determinant theory [11, 9, 12, 8]. Our use of $A_{n-1}$, lattice saturation, and Smith normal form follows standard lattice and integral-matrix language [5, 14]. We treat a Sudoku grid as a linear-algebraic object parameterized by symbol relabelings, and study the odd-rank locus, its kernel certificates in $A_{n-1}$, and the algebra of its stabilizers under the natural $S_n$-action; none of the works above address these questions.

This paper is also deliberately narrower than several nearby problems. It does not count Sudoku grids, solve completion or clue-minimality problems, classify autotopism groups, or assert that the laboratory base $M_{19}$ is extremal among all order-9 Sudoku grids. Its contribution is the rank-and-kernel geometry of the centered relabeling family and the reproducible identification of the finite structures that arise from that geometry.

Within the present project, two sibling papers form the immediate context. The companion paper *Determinant Divisibility of Centered Latin Squares* [1] establishes the $\det(\widehat{E}) = n\det(A)$ identity and divisibility consequences for the centered operator; its results are used here as a base layer and not re-proved. The working draft *Rank-parity falsification of Latin square operators* [2] provides the existence of odd-rank examples; the present paper goes from existence to algebraic structure (kernel lattice, stabilizer groups, affine coset symmetry, autotopism rigidity).

### 1.6 Organization

Section 2 introduces the centered operator $E_{\gamma,L}$, the standard hyperplane, and exact-arithmetic rank/kernel computation. Section 3 proves the Box Band Lemma. Section 5 treats lower-order calibration at $n = 6$ and $n = 7$. Sections 6–7 establish the lift lemma and $A_8$-saturation. Section 8 identifies the affine coset-symmetry group of $M_{19}$. Section 9 contains the stabilizer dichotomy. Section 10 proves Latin- square rigidity of $L_{M_{19}}$ and its consequences. Section 11 describes the computational methodology. Section 12 lists open problems and out-of-scope directions. Appendices A–D collect the certified data.

# 2 Centered Sudoku operators

## 2.1 Sudoku grids as constrained Latin squares

A *Latin square* of order $n$ is an $n \times n$ array $L = (L_{ij})_{1 \le i,j \le n}$ with $L_{ij} \in \{1, \dots, n\}$ such that every symbol appears exactly once in each row and once in each column. A *Sudoku grid* of order $n = m^2$ is a Latin square that in addition satisfies the box constraint: the $m \times m$ blocks $\{(i, j) : (I-1)m < i \le Im,\ (J-1)m < j \le Jm\}$, indexed by $(I, J) \in \{1, \dots, m\}^2$, each contain every symbol of $\{1, \dots, n\}$ exactly once. The classical case is $n = 9$, $m = 3$; the lower-order calibration in §5 also uses $n = 6$ (the $2 \times 3$ Roku-Doku family) and $n = 7$ (non-Sudoku Latin squares used as a comparison class).

We write $\mathrm{LS}(n)$ for the set of Latin squares of order $n$ and $\mathrm{Sud}(n) \subseteq \mathrm{LS}(n)$ for the subset of square Sudoku grids when $n$ is a perfect square. More generally, when $n = hw$ we write $\mathrm{Sud}_{h,w}(n)$ for the rectangular $h \times w$ Sudoku/Roku-Doku family. Thus the order-6 calibration uses $\mathrm{Sud}_{2,3}(6)$, not the square-grid notation $\mathrm{Sud}(6)$.

Throughout the paper, $D_n$ denotes the geometric dihedral group of order $2n$; thus $D_6$ has order 12 and $D_4$ has order 8.

## 2.2 Symbol relabelings $\gamma \in S_n$

The symmetric group $S_n$ acts on $\mathrm{LS}(n)$ by *symbol relabeling*: for $\gamma \in S_n$ and $L \in \mathrm{LS}(n)$,

$$(\gamma(L))_{ij} \ := \ \gamma(L_{ij}). \tag{3}$$

This action stabilizes $\mathrm{Sud}(n)$, when defined, and similarly stabilizes $\mathrm{Sud}_{h,w}(n)$; it commutes with the row, column, band, and stack permutations of the standard Latin-square isotopy group: $\gamma$ relabels the alphabet but preserves the combinatorial frame. In particular, an autotopism of $L$ in the sense of [16, 6] is a triple $(\alpha, \beta, \gamma) \in S_n^3$ acting on rows, columns, and symbols with $(\alpha, \beta, \gamma) \cdot L = L$; the present paper isolates the third coordinate.

## 2.3 Centered matrix $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2} J_n$

Treat $\gamma(L) \in \mathbb{Z}^{n \times n}$ as an integer matrix with entries in $\{1, \dots, n\}$. The *centered operator* of the pair $(L, \gamma)$ is the rational matrix

$$E_{\gamma,L} \ := \ \gamma(L) \ - \ \frac{n+1}{2} \, J_n \ \in \ \mathbb{Q}^{n \times n}, \tag{4}$$

where $J_n = \mathbf{1}\mathbf{1}^\top$ is the all-ones matrix and $\frac{n+1}{2}$ is the symbol mean. Each row of $\gamma(L)$ is a permutation of $\{1, \dots, n\}$, hence has constant sum $\frac{n(n+1)}{2}$; the same holds for each column. Subtracting the all-ones rank-one matrix $\frac{n+1}{2} J_n$ centers both row and column sums, giving

$$E_{\gamma,L} \, \mathbf{1} \ = \ 0, \qquad \mathbf{1}^\top E_{\gamma,L} \ = \ 0. \tag{5}$$

The centering is identical to that of the sibling determinant paper [1], which studies $\det(\widehat{E}_{\gamma,L})$; here we focus on $\operatorname{rank} E_{\gamma,L}$ and the structure of its kernel certificates.

### 2.4 The standard subspace $V_{\mathrm{std},n}$ and certificate vectors

Let

$$V_{\mathrm{std},n} \;:=\; \{x \in \mathbb{Q}^n : \mathbf{1}^\top x = 0\} \tag{6}$$

be the standard hyperplane. By (5) the operator $E_{\gamma,L}$ maps $\mathbb{Q}^n$ into $V_{\mathrm{std},n}$ and annihilates the line $\mathbb{Q}\mathbf{1}$, hence restricts to an endomorphism $E_{\gamma,L} : V_{\mathrm{std},n} \to V_{\mathrm{std},n}$. Choose the basis $\{e_1 - e_n, \dots, e_{n-1} - e_n\}$ of $V_{\mathrm{std},n}$ and let $P_n \in \mathbb{Z}^{n\times(n-1)}$ be the corresponding inclusion matrix. Define

$$\widehat{E}_{\gamma,L} \;:=\; P_n^\top\, E_{\gamma,L}\, P_n \;\in\; \mathbb{Q}^{(n-1)\times(n-1)}. \tag{7}$$

Then $\widehat{E}_{\gamma,L}$ is the matrix of $E_{\gamma,L}|_{V_{\mathrm{std},n}}$ in the chosen basis and

$$\operatorname{rank} E_{\gamma,L} \;=\; \operatorname{rank} \widehat{E}_{\gamma,L} \;\in\; \{0, 1, \dots, n-1\}. \tag{8}$$

*Remark* 2.1 ($\widehat{E}_{\gamma,L}$ as a Gram-projected coordinate matrix). Strictly, the matrix $\widehat{E}_{\gamma,L} = P_n^\top E_{\gamma,L} P_n$ in (7) is the matrix of $E_{\gamma,L}|_{V_{\mathrm{std},n}}$ with respect to the dual basis specified by $P_n$ on the left and the chosen basis of $V_{\mathrm{std},n}$ on the right; equivalently, $P_n^\top P_n$ acts as the Gram matrix of the basis $\{e_i - e_n\}_{i<n}$, which is invertible ($\det(P_n^\top P_n) = n$). Hence the rank of $\widehat{E}_{\gamma,L}$ coincides with the rank of the abstract restricted endomorphism $E_{\gamma,L}|_{V_{\mathrm{std},n}} : V_{\mathrm{std},n} \to V_{\mathrm{std},n}$, justifying (8).

We say $(L, \gamma)$ is *full-rank* if $\operatorname{rank} \widehat{E}_{\gamma,L} = n - 1$ and *odd-rank* if $\operatorname{rank} \widehat{E}_{\gamma,L} = n - 2$. The term "odd-rank" is used for the rank-drop-one locus in the odd-order cases studied below; the order-6 discussion is a separate even-characteristic calibration. The intermediate root lattice

$$A_{n-1} \;:=\; V_{\mathrm{std},n} \cap \mathbb{Z}^n \tag{9}$$

is the integer lattice on which kernel certificates of odd-rank relabelings will be exhibited (§6). Notation conventions, in particular the reconciliation with $\widehat{E} = P^\top E P$ used in the sibling paper, are collected in `certified/SIBLING_NOTATION.md`.

*Remark* 2.2 (Canonical rank vs. relabeling rank). At $\gamma = \mathrm{id}$ the operator $E_{\mathrm{id},L} = L - \frac{n+1}{2} J_n$ is sometimes called the *canonical* centered matrix of $L$ and its rank the *canonical rank*. A canonical-rank scan over 400 Sudoku grids sampled at `BASE_SEED=20260440` returns the histogram $\{n-1\colon 400\}$; no sample grid is canonically odd-rank [2]. Thus the rank-$(n-2)$ phenomena studied below are produced by relabeling $\gamma \neq \mathrm{id}$ and not by the underlying Sudoku grid alone.

### 2.5 Left and right kernel loci

For a fixed Latin square $L$ and vectors $w, u \in V_{\mathrm{std},n}$, define the raw left and right kernel loci

$$\Gamma_L(L, w) \;:=\; \{\gamma \in S_n : w^\top E_{\gamma,L} = 0\}, \tag{10}$$
$$\Gamma_R(L, u) \;:=\; \{\gamma \in S_n : E_{\gamma,L} u = 0\}. \tag{11}$$

These loci are not assumed to be groups; in the examples below they are usually finite subsets of $S_n$ which may become cosets after a choice of basepoint.

**Lemma 2.3** (Transpose duality for kernel loci)**.** *For every Latin square $L$ and every $u \in V_{\mathrm{std},n}$,*

$$\Gamma_R(L, u) \;=\; \Gamma_L(L^\top, u). \tag{12}$$

*Proof.* Symbol relabeling commutes with transpose, so $\gamma(L^\top) = \gamma(L)^\top$ and hence $E_{\gamma,L^\top} = E_{\gamma,L}^\top$. Therefore $E_{\gamma,L} u = 0$ if and only if $u^\top E_{\gamma,L}^\top = 0$, which is precisely the condition $\gamma \in \Gamma_L(L^\top, u)$. □

### 2.6 Exact rank and kernel computation

All rank, kernel, and Smith-normal-form computations in this paper are exact-arithmetic over $\mathbb{Q}$ or $\mathbb{Z}$. The pipeline used by the certified scripts in `certified/` combines:

(i) the Bareiss fraction-free elimination algorithm [3] for integer determinants and ranks of $E_{\gamma,L}$ and $\widehat{E}_{\gamma,L}$;

(ii) rational row reduction (RREF over $\mathbb{Q}$) via SymPy for nullspace bases of $\widehat{E}_{\gamma,L}$;

(iii) Smith normal form over $\mathbb{Z}$ for the elementary divisors of $E_{\gamma,L}$ and of the certificate sublattices;

(iv) finite-field cross-checks at $p \in \{2, 3, 5, 7\}$ for independent verification of integer ranks.

Each odd-rank relabeling reported in this paper has been verified by all four routes; §11 gives the cross-check ledger.

## 3 The Sudoku Box Band Lemma

We now formalize the linear identity (2) announced in the introduction. Throughout this section $n = m^2$, $L \in \mathrm{LS}(n)$, and $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2} J_n$.

### 3.1 Row-band and column-stack averaging maps

For $n = m^2$, partition the $n$ rows of an $n \times n$ matrix into $m$ *row-bands* of size $m$ each:

$$R_I \;:=\; \{(I-1)m + 1, \dots, Im\}, \qquad I \in \{1, \dots, m\},$$

and analogously partition the columns into $m$ *column-stacks* $C_J$, $J \in \{1, \dots, m\}$. Let $B_{\mathrm{rb}} \in \{0,1\}^{n\times m}$ have $(B_{\mathrm{rb}})_{iI} = 1$ iff $i \in R_I$, and similarly $B_{\mathrm{cs}} \in \{0,1\}^{n\times m}$ for the column-stacks. The matrices

$$P_{\mathrm{rb}} \;:=\; \frac{1}{m}\, B_{\mathrm{rb}}^{\top}, \qquad P_{\mathrm{cs}} \;:=\; \frac{1}{m}\, B_{\mathrm{cs}}^{\top} \tag{13}$$

average the entries of a vector over row-bands and column-stacks respectively. We use both the unnormalized form $B_{\mathrm{rb}}^{\top} M B_{\mathrm{cs}}$ and the normalized form $P_{\mathrm{rb}} M P_{\mathrm{cs}}^{\top}$; they differ by the scalar $m^{-2}$ and vanish simultaneously.

### 3.2 Statement and proof

**Theorem 3.1** (Box Band Lemma)**.** *Let $n = m^2$, $L \in \mathrm{Sud}(n)$, and $\gamma \in S_n$. Then*

$$B_{\mathrm{rb}}^{\top}\, E_{\gamma,L}\, B_{\mathrm{cs}} \;=\; 0 \;\in\; \mathbb{Z}^{m\times m}, \tag{14}$$

*or equivalently $P_{\mathrm{rb}}\, E_{\gamma,L}\, P_{\mathrm{cs}}^{\top} = 0$.*

*Proof.* Fix a row-band index $I$ and a column-stack index $J$. The Sudoku box constraint applied to the $(I, J)$ box says that the multiset $\{\gamma(L)_{ij} : i \in R_I,\ j \in C_J\}$ equals the multiset $\{1, \dots, n\}$, since $\gamma$ is a bijection of $\{1, \dots, n\}$ and $L$ is Sudoku. Hence

$$\sum_{i\in R_I} \sum_{j\in C_J} \gamma(L)_{ij} \;=\; \sum_{k=1}^{n} k \;=\; \frac{n(n+1)}{2}.$$

On the other hand,

$$\sum_{i\in R_I} \sum_{j\in C_J} \frac{n+1}{2} \;=\; m^2 \cdot \frac{n+1}{2} \;=\; \frac{n(n+1)}{2},$$

since $|R_I| = |C_J| = m$ and $m^2 = n$. Therefore

$$(B_{\mathrm{rb}}^\top E_{\gamma,L} B_{\mathrm{cs}})_{IJ} \;=\; \sum_{i\in R_I}\sum_{j\in C_J} (E_{\gamma,L})_{ij} \;=\; \frac{n(n+1)}{2} - \frac{n(n+1)}{2} \;=\; 0,$$

which proves (14). □

### 3.3 A cyclic Latin-square witness

The Sudoku hypothesis is essential. Let $L^{\mathrm{cyc}}$ denote the cyclic Latin square of order 9 with $L^{\mathrm{cyc}}_{ij} \equiv i + j - 1 \pmod 9$, restored to $\{1, \dots, 9\}$. This is a Latin square but not a Sudoku grid: the $(1, 1)$ box reads $\{1, 2, 3, 2, 3, 4, 3, 4, 5\}$, which is not a permutation of $\{1, \dots, 9\}$. A direct evaluation gives

$$B_{\mathrm{rb}}^\top \, E_{\mathrm{id},L^{\mathrm{cyc}}} \, B_{\mathrm{cs}} \;=\; \begin{pmatrix} -18 & 9 & 9 \\ 9 & 9 & -18 \\ 9 & -18 & 9 \end{pmatrix}, \tag{15}$$

with squared Frobenius norm $1458 \neq 0$. The value 1458 is the Frobenius norm squared on this specific cyclic instance and is not claimed to be extremal across the family of order-9 Latin squares that fail the Sudoku hypothesis; we cite it only to certify that the identity (14) is not a vacuous consequence of Latin-squareness. The same matrix, evaluated on the Sudoku grids `base1` and $M_{19}$ (defined in §4 and §8), is identically zero. The witness triple `(M19, base1, cyclic-LS9)` is contained in the certified file `box_band_lemma_witness.json`; the Sudoku hypothesis cannot be replaced by the Latin-square hypothesis without breaking (14).

### 3.4 Rectangular and gerechte variants

The proof of Theorem 3.1 does not use equality of box height and width except in the notation. It extends verbatim to rectangular Sudoku boxes.

**Corollary 3.2** (Rectangular Box Band Lemma)**.** *Let $n = hw$, and let $L$ be a Latin square of order $n$ equipped with a rectangular Sudoku structure whose boxes are the Cartesian products of row-bands of height $h$ and column-stacks of width $w$, each box containing every symbol exactly once. Let $B_{\mathrm{rb}}$ be the $n \times w$ row-band incidence matrix and $B_{\mathrm{cs}}$ the $n \times h$ column-stack incidence matrix. Then for every $\gamma \in S_n$,*

$$B_{\mathrm{rb}}^\top E_{\gamma,L} B_{\mathrm{cs}} \;=\; 0 \;\in\; \mathbb{Z}^{w\times h}. \tag{16}$$

*Equivalently, the corresponding normalized averaging maps also give zero.*

*Proof.* Each entry of the matrix in (16) is the centered sum over one $h \times w$ box. The uncentered sum is $\sum_{k=1}^{n} k = n(n+1)/2$, and the centering contribution is $hw \cdot (n+1)/2 = n(n+1)/2$. □

For general gerechte designs the correct statement is a region-sum identity rather than a row-averaging/column-averaging factorization.

**Lemma 3.3** (Gerechte region-sum identity)**.** *Let $L$ be a Latin square of order $n$, and let $\mathcal{R} = \{R_1, \dots, R_n\}$ be a partition of the $n^2$ cells into $n$ regions of size $n$ such that every region contains each symbol exactly once. Then for every $\gamma \in S_n$ and every region $R_a \in \mathcal{R}$,*

$$\sum_{(i,j)\in R_a} (E_{\gamma,L})_{ij} \;=\; 0. \tag{17}$$

*If the regions are Cartesian products of a row partition and a column partition, this is exactly the rectangular Box Band Lemma. For a non-Cartesian gerechte partition there need not be a matrix identity of the form $B_{\mathrm{row}}^\top E_{\gamma,L} B_{\mathrm{col}} = 0$; the invariant lives on cell-region incidence instead.*

*Proof.* For a fixed region $R_a$, the multiset $\{\gamma(L_{ij}) : (i,j) \in R_a\}$ is $\{1,\ldots,n\}$, since $R_a$ contains every symbol once and $\gamma$ is a bijection. Thus the uncentered regional sum is $n(n+1)/2$, exactly equal to the sum of the centering term $(n+1)/2$ over the $n$ cells of $R_a$. $\square$

The certified witness file `box_band_lemma_witness.json` checks the square $3 \times 3$ Sudoku case, the rectangular $2 \times 3$ Roku-Doku case, and a non-Cartesian gerechte partition of the cyclic Latin square of order 9 by broken diagonals.

**Corollary 3.4** (Box-band kernel inclusion)**.** *For every Sudoku grid $L$ and every $\gamma \in S_n$, the column span of $B_{\mathrm{cs}}$ is contained in the right kernel of $B_{\mathrm{rb}}^{\top} E_{\gamma,L}$, and the column span of $B_{\mathrm{rb}}$ is contained in the left kernel of $E_{\gamma,L} B_{\mathrm{cs}}$.*

*Proof.* Immediate from (14). $\square$

Theorem 3.1 is the structural input that distinguishes Sudoku from the Latin-square axioms alone at the linear-algebraic level. The remainder of the paper exploits the fact that the box constraint is *linear* (after centering) to import lattice and group-theoretic machinery to the analysis of $E_{\gamma,L}$.

# 4 Odd-rank relabelings of $\mathbf{base}_1$

We now turn from structural identities to enumerative facts. The goal of this section is to describe the rank distribution of the family $\{E_{\gamma,L}\}_{\gamma \in S_9}$ on a single, fixed Sudoku grid `base1`, and to certify the resulting odd-rank relabelings by four independent exact-arithmetic routes.

## 4.1 The Sudoku base `base1`

The grid `base1` $\in \mathrm{Sud}(9)$ used throughout this section is the canonical $9 \times 9$ grid

$$\texttt{base1} = \begin{pmatrix} 5 & 3 & 4 & 6 & 7 & 8 & 9 & 1 & 2 \\ 6 & 7 & 2 & 1 & 9 & 5 & 3 & 4 & 8 \\ 1 & 9 & 8 & 3 & 4 & 2 & 5 & 6 & 7 \\ 8 & 5 & 9 & 7 & 6 & 1 & 4 & 2 & 3 \\ 4 & 2 & 6 & 8 & 5 & 3 & 7 & 9 & 1 \\ 7 & 1 & 3 & 9 & 2 & 4 & 8 & 5 & 6 \\ 9 & 6 & 1 & 5 & 3 & 7 & 2 & 8 & 4 \\ 2 & 8 & 7 & 4 & 1 & 9 & 6 & 3 & 5 \\ 3 & 4 & 5 & 2 & 8 & 6 & 1 & 7 & 9 \end{pmatrix}. \tag{18}$$

This is the same grid recovered, byte-equal, by the audit script `recover_base1_22_perms.py` and recorded in the certified package.

## 4.2 Exhaustive $S_9$ scan

For each of the $|S_9| = 362{,}880$ symbol relabelings $\gamma$, the rank of $E_{\gamma,\texttt{base1}}$ on $V_{\mathrm{std},n}$ is computed by exact integer Bareiss elimination. The resulting rank distribution is stated in Certified Proposition 4.1.

**Certified Proposition 4.1** (Odd-rank scan on `base1`)**.** *On the Sudoku base `base1` of order $n = 9$, the rank distribution of the family $\{\widehat{E}_{\gamma,L}\}_{\gamma \in S_9}$ is*

$$\#\{\gamma \in S_9 : \operatorname{rank} \widehat{E}_{\gamma,L} = r\} = \begin{cases} 362{,}858, & r = 8, \\ 22, & r = 7, \\ 0, & r \notin \{7,8\}. \end{cases} \tag{19}$$

*The 22 odd-rank relabelings are recorded explicitly in the corresponding certificate.*

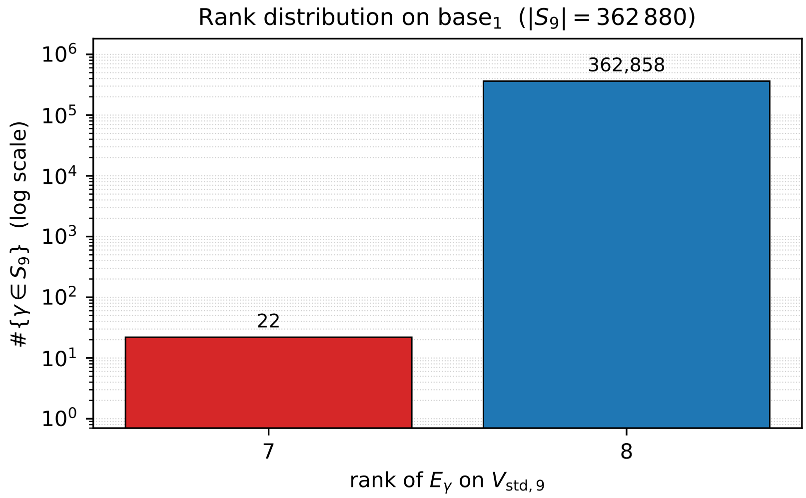


Figure 1: Rank distribution of $\{\widehat{E}_{\gamma,L}\}_{\gamma\in S_9}$ on `base1`. Logarithmic vertical axis. The two non-empty bins have populations 362,858 (rank 8) and 22 (rank 7); no $\gamma \in S_9$ produces rank $\widehat{E}_{\gamma,L} \leq 6$.

This is an EXHAUSTIVE result: the entire group $S_9$ has been enumerated, no $\gamma$ has been omitted, and the rank of each $\widehat{E}_\gamma$ has been recorded. Figure 1 displays the histogram on a logarithmic scale.

### 4.3 Four-way exact certification

Each of the 22 odd-rank relabelings has been independently certified by four exact-arithmetic procedures:

(i) **Integer rank** via Bareiss elimination on $E_{\gamma,L} \in \mathbb{Z}^{9\times 9}$ followed by Gram projection to $\widehat{E}_{\gamma,L} \in \mathbb{Z}^{8\times 8}$;

(ii) **Rational kernel** via SymPy nullspace of $\widehat{E}_{\gamma,L}$, yielding a one-dimensional kernel $\mathbb{Q}$-vector for every $\gamma$ in the list, with all entries in $\mathbb{Z}$ after a uniform denominator clear-out;

(iii) **Integer Smith normal form** of $E_{\gamma,L} \in \mathbb{Z}^{9\times 9}$. The elementary-divisor diagonal is rank-7 on every $\gamma$ in the list (seven nonzero invariants and two trailing zeros), but the seventh invariant is *not* uniformly 1: across the 22 relabelings five distinct diagonals occur, namely

$$\begin{array}{ll} (1,1,1,1,1,1,1,0,0) & \text{on 10 perms,} \\ (1,1,1,1,1,1,2,0,0) & \text{on 4 perms,} \\ (1,1,1,1,1,1,4,0,0) & \text{on 4 perms,} \\ (1,1,1,1,1,1,3,0,0) & \text{on 2 perms,} \\ (1,1,1,1,1,2,10,0,0) & \text{on 2 perms.} \end{array} \tag{20}$$

In particular only 10/22 relabelings have torsion-free cokernel on $\mathbb{Z}^9/\mathrm{im}(E_{\gamma,L})$. Among the remaining 12, the torsion factors are $\mathbb{Z}/2$, $\mathbb{Z}/4$, $\mathbb{Z}/3$, and, for the final SNF class, $\mathbb{Z}/2 \oplus \mathbb{Z}/10$. The closed-form description of this distribution as a function of $\gamma$ is recorded as open problem (O1) in §12.2;

(iv) **$\mathbb{F}_2$-Smith normal form** of the reduction of $E_{\gamma,L}$ modulo 2. The integer rank 7 does *not* lift uniformly to $\mathbb{F}_2$: the rank distribution over the 22 relabelings is $\{\mathrm{rank}_{\mathbb{F}_2} = 7 : 12,\ 6 : 8,\ 5 : 2\}$. Twelve relabelings agree mod 2 (rank 7); ten drop in rank, consistent with the $\mathbb{Z}$-torsion at 2 in part (iii).

The four routes are independent in the sense that no two of them share a common code path past the construction of $E_{\gamma,L}$. Their agreement on the 22-element list (rank 7 over $\mathbb{Z}$, kernel 1-dimensional in $V_{\mathrm{std}}$, integer SNF in the five-class spectrum (20), and the $\mathbb{F}_2$ rank histogram above) rules out single-bug, single-route, and single-prime artefacts. The per-perm certificates are stored in `four_way_base1_22.json` and summarized in Appendix A.

### 4.4 Scope caveat

Certified Proposition 4.1 concerns *one* Sudoku base under *all* symbol relabelings. It is *not* a distributional statement over Sud(9). Comparable corpus-level statements require either an exhaustive enumeration of Sud(9) (out of reach: $|\mathrm{Sud}(9)| \approx 6.67 \times 10^{21}$ in the sense of [10]) or a sampling caveat (cf. §11). In particular, the count $22/362{,}880 \approx 6.06 \times 10^{-5}$ is the odd-rank rate for `base1`, not for the population of Sudoku grids.

## 5 Lower-order calibration

The exhaustive analysis of `base1` at $n = 9$ is one data point in Sud(9). To calibrate it against smaller orders, this section records two facts at $n \in \{6, 7\}$ that together motivate the structural treatment of §§6–8: (i) the box constraint does not suppress the $\mathbb{F}_2$-rank parity proxy at $n = 6$ (i.e. the rate of grids for which the determinant-paper auxiliary matrix has full $\mathbb{F}_2$-rank is comparable on rectangular Sudoku and on all Latin squares of order 6), and (ii) at $n = 7$ the symmetry group of the simplest odd-rank certificate is dihedral of order 12 and embeds inside the signed-pair group $B_3 < S_7$.

**Lemma 5.1** (Even-characteristic projected-rank ceiling)**.** *Let $n$ be even, let $L$ be any Latin square of order $n$, and reduce entries modulo 2. With $P_n = (e_i - e_n)_{i<n}$ as in (7), the Gram-projected matrix*

$$P_n^\top L P_n \;\in\; \mathbb{F}_2^{(n-1)\times(n-1)}$$

*has rank at most $n - 2$ over $\mathbb{F}_2$.*

*Proof.* In characteristic 2, subtraction and addition coincide, so $P_n \mathbf{1}_{n-1} = \mathbf{1}_n$ because $n - 1$ is odd. Since each row of a Latin square has the same symbol sum, $L\mathbf{1}_n = s\mathbf{1}_n$ over $\mathbb{F}_2$ for some $s \in \mathbb{F}_2$. But $P_n^\top \mathbf{1}_n = 0$, again because each column of $P_n$ has two nonzero entries in characteristic 2. Therefore

$$(P_n^\top L P_n)\mathbf{1}_{n-1} = P_n^\top L\mathbf{1}_n = sP_n^\top \mathbf{1}_n = 0.$$

The vector $\mathbf{1}_{n-1}$ is nonzero, so the projected matrix has a nontrivial kernel. □

Lemma 5.1 applies to the Gram-projected matrix $P_n^\top L P_n$ in characteristic 2. It is separate from the determinant-paper auxiliary matrix $A_L$ used in the census below, which can have rank $n - 1$.

### 5.1 Roku-Doku at $n = 6$: $\mathbb{F}_2$-rank census

For $n = 6$, with first row fixed to $(1, 2, 3, 4, 5, 6)$, the finite corpora have sizes

$$|\mathrm{LS}(6)|_{\text{first row fixed}} = 1{,}128{,}960, \qquad |\mathrm{Sud}_{2,3}(6)|_{\text{first row fixed}} = 39{,}168.$$

The statistic used in this calibration is not rank $\widehat{E}_{\gamma,L}$ over $\mathbb{F}_2$. Instead, for each square $L$ in either family, we use the determinant-paper auxiliary matrix

$$A_L \;:=\; (L_{ij} - L_{i,6})_{1\le i,j\le 5} \quad \text{over } \mathbb{F}_2,$$

the rank-parity proxy of [2]. We record the proportion of witnesses for which $A_L$ has full $\mathbb{F}_2$-rank, namely $\mathrm{rank}_{\mathbb{F}_2} A_L = n - 1 = 5$.

**Certified Proposition 5.2** (Roku-Doku exhaustive census)**.** *With the first row fixed to* $(1, 2, 3, 4, 5, 6)$,

$$\#\{L \in \mathrm{Sud}_{2,3}(6) : \mathrm{rank}_{\mathbb{F}_2} A_L = 5\} = 2{,}592 \quad (=6.6177\%),$$
$$\#\{L \in \mathrm{LS}(6) : \mathrm{rank}_{\mathbb{F}_2} A_L = 5\} = 69{,}120 \quad (=6.1224\%).$$

The census is exhaustive on the two fixed-first-row finite corpora; no sampling inference is used. We phrase the conclusion conservatively. The box constraint does *not* suppress the full-rank-over-$\mathbb{F}_2$ phenomenon for the auxiliary matrix $A_L$ at $n = 6$; in the exhaustive Roku-Doku census, the proportion of full-rank-over-$\mathbb{F}_2$ witnesses is slightly larger than in the Latin-square census. The full rank distribution at $n = 6$ across the five strata $\{1, 2, 3, 4, 5\}$ is reported in `n6_odd_rank_census.json`.

## 5.2 The $n = 7$ group $H_{V_1} \cong D_6$

For $n = 7$, fix the laboratory Latin square $L_{V_1}$ of order 7 recorded in `scripts/phase_9_15.py`, and let $B_3 = C_2 \wr S_3 \subset S_7$ be the signed-pair group of order 48: it permutes the three pairs $\{1, 7\}$, $\{2, 6\}$, $\{3, 5\}$, may swap the two entries inside each pair, and fixes 4. Define

$$H_{V_1} \;:=\; \{\gamma \in B_3 \;:\; \mathrm{rank}\,\widehat{E}_{\gamma,L_{V_1}} \;=\; 5\}, \tag{21}$$

the intersection of the odd-rank symbol-relabeling locus of $L_{V_1}$ with $B_3$. Equivalently, $H_{V_1}$ is the *rank-locus subset* of $B_3$ for the base $L_{V_1}$.

*Remark* 5.3 ($H_{V_1}$ is not a coordinate stabilizer)*.* Any vector $v \in \mathbb{R}^7$ with all-distinct coordinates (such as a linear template $(-3, -2, -1, 0, 1, 2, 3) \in A_6$) has trivial coordinate stabilizer in $S_7$. Hence $H_{V_1}$ cannot be defined as the coordinate stabilizer of any such vector; it is intrinsically a *rank-locus* object, defined by the operator condition (21) on $\widehat{E}_{\gamma,L_{V_1}}$.

**Certified Proposition 5.4** (Stabilizer at $n = 7$)**.** *The set $H_{V_1}$ defined by (21) satisfies $|H_{V_1}| = 12$, is closed under composition (hence a subgroup of $B_3$), and is isomorphic to the dihedral group $D_6 \cong S_3 \times \mathbb{Z}/2$. Explicit generators are*

$$r \;=\; (3, 1, 6, 4, 2, 7, 5) \in B_3, \qquad s \;=\; (6, 7, 3, 4, 5, 1, 2) \in B_3,$$

*satisfying $r^6 = s^2 = (rs)^2 = e$ and generating $H_{V_1}$ in full. The rank distribution of $\widehat{E}_{\gamma,L_{V_1}}$ on $B_3$ is $\{\mathrm{rank}\,5 : 12,\ \mathrm{rank}\,6 : 36\}$; the rank-5 rate inside $B_3$ is $12/48 = 1/4$, against a baseline rate $30/5040 = 1/168$ in $S_7$, an enrichment ratio of 42.*

*Certification.* The exhaustive enumeration of rank $\widehat{E}_{\gamma,L_{V_1}}$ for $\gamma \in B_3$ is recorded in the certificate `n7_HV1_D6.json`; this directly gives $|H_{V_1}| = 12$. Closure under composition is checked element-by-element on the 12-element list. The element-order multiset of $H_{V_1}$ is $\{1{:}1,\ 2{:}7,\ 3{:}2,\ 6{:}2\}$, which uniquely identifies $D_6$ among groups of order 12: the alternatives $\mathbb{Z}/12$, $\mathbb{Z}/6 \times \mathbb{Z}/2$, $A_4$, and $\mathrm{Dic}_3$ all have different order multisets, as recorded in the same certificate. Constructively, $r$ has order 6, $s$ has order 2, and $srs = r^{-1}$ holds in $S_7$ by direct computation; hence $\langle r, s\rangle = D_6$. The order-12 subgroup $\langle r, s\rangle$ coincides with $H_{V_1}$ on enumeration. Membership $r, s \in B_3$ is verified by checking that each preserves the pair system $\{1, 7\} \cup \{2, 6\} \cup \{3, 5\}$ and fixes 4; this is implemented in `check_b3_subgroup.py`. □

The Cayley diagram of $\langle r, s\rangle$ is shown in Figure 2; it is hexagonal-prismatic, with the red 6-cycle $r$ generating the rotational subgroup and the blue involution $s$ generating the reflection.

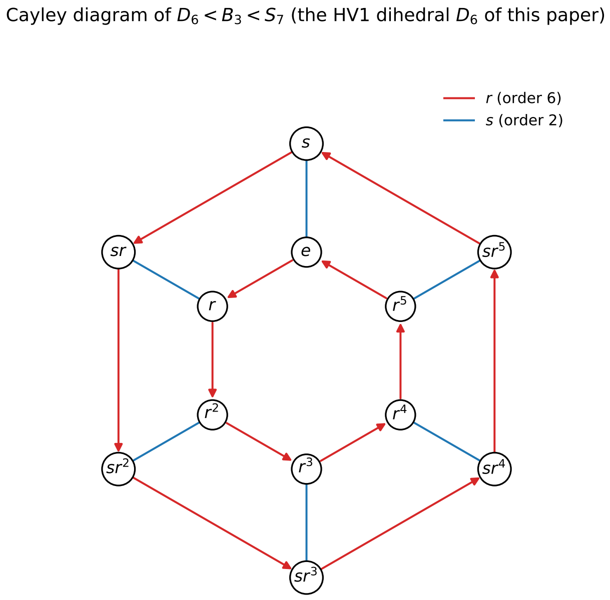


Figure 2: Cayley diagram of the rank-locus subgroup $H_{V_1} \cong D_6$ at $n = 7$. Red edges: generator $r$ of order 6; blue edges: reflection $s$ of order 2.

### 5.3 Bridge to rank-parity falsification

Certified Proposition 4.1 falsifies the rank-parity statement "rank $\widehat{E}_{\gamma,L} \equiv n-1 \pmod 2$ for every Sudoku grid $L$ and every $\gamma$": `base1` admits 22 relabelings with even-corank (odd-rank) $\widehat{E}_{\gamma,L}$. Certified Proposition 5.2 records a separate lower-order calibration for the determinant-paper auxiliary $\mathbb{F}_2$ proxy at rectangular order 6; it is not an additional proof about rank $\widehat{E}_{\gamma,L}$ over $\mathbb{F}_2$. These finite counterexamples and calibrations are certified inside the present repository and are the only rank-parity input used below. The broader rank-parity falsification program is recorded in the sibling working draft [2]; here we proceed from the certified counterexamples to their algebraic structure (§§6–10).

## 6 Certificates in $V_{\mathrm{std},n}$

For every odd-rank pair $(L, \gamma)$, choose a nonzero $w \in \ker \widehat{E}_{\gamma,L}$. Lifting $w \in \mathbb{Q}^{n-1}$ along $P_n$ gives a vector $w \in V_{\mathrm{std},n} \cap \mathbb{Q}^n$, and after clearing denominators a vector in $A_{n-1} = V_{\mathrm{std},n} \cap \mathbb{Z}^n$. We refer to such an integer representative as a *kernel certificate* of $(L, \gamma)$. The purpose of this section is twofold: first, the Lift Lemma (Theorem 6.1) states that the kernel certificate is the same whether read off $E_{\gamma,L}$ or off $\gamma(L)$; second, the support and height taxonomy (§6.2) classifies the kernel certificates that arise in practice.

### 6.1 The Lift Lemma

**Theorem 6.1** (Lift Lemma)**.** *Let $L \in \mathrm{LS}(n)$, $\gamma \in S_n$, and $w \in V_{\mathrm{std},n}$. Then*

$$E_{\gamma,L}\, w \;=\; 0 \quad\Longleftrightarrow\quad \gamma(L)\, w \;=\; 0, \tag{22}$$

*and dually for left kernels: $w^\top E_{\gamma,L} = 0 \iff w^\top \gamma(L) = 0$.*

*Proof.* For any $w \in V_{\mathrm{std},n}$ we have $\mathbf{1}^\top w = 0$, hence $J_n w = \mathbf{1}(\mathbf{1}^\top w) = 0$. Therefore

$$E_{\gamma,L} w \;=\; \gamma(L)\, w \;-\; \frac{n+1}{2} J_n w \;=\; \gamma(L)\, w,$$

so the two expressions $E_{\gamma,L}w$ and $\gamma(L)w$ are identically equal on $V_{\mathrm{std},n}$. The biconditional is immediate, and the dual statement follows from $w^\top J_n = 0$. $\square$

**Corollary 6.2** (Kernel certificates lift to $A_{n-1}$)**.** *For every odd-rank pair $(L, \gamma)$, the right kernel of $\widehat{E}_{\gamma,L}$ is one-dimensional and lifts to a one-dimensional subspace of $V_{\mathrm{std},n}$. A primitive integer generator of that lift lies in $A_{n-1}$.*

The integrality of the primitive generator is automatic: the kernel is a $\mathbb{Q}$-line $\mathbb{Q}v \subseteq V_{\mathrm{std},n}$; clearing denominators in $v$ and dividing by the gcd of its entries yields a primitive integer vector $w \in \mathbb{Z}^n \cap V_{\mathrm{std},n} = A_{n-1}$ (with $\gcd_i w_i = 1$), unique up to sign.

The corollary is verified independently on the 22 `base1` relabelings of §4 and on the 100 relabelings of $M_{19}$ scanned in §8; the relevant histograms are

$$\dim_{\mathbb{Q}} \ker E_{\gamma,L}|_{\mathbb{Q}^9} = 2 \text{ (22 times for base}_1\text{, 100 times for } M_{19}\text{)},$$

$$\dim_{\mathbb{Q}} \ker \widehat{E}_{\gamma,L}|_{V_{\mathrm{std},n}} = 1 \text{ (every odd-rank case)},$$

recorded in `lift_lemma_evidence.json`. The doubled ambient kernel dimension reflects the fact that $\mathbb{Q}\mathbf{1} \subset \ker E_{\gamma,L}$ unconditionally; the lift to $V_{\mathrm{std},n}$ removes that trivial line.

### 6.2 Support and height taxonomy

A kernel certificate $w \in A_{n-1}$ is classified by two coarse invariants:

- the *support* $\mathrm{supp}(w) = \{i : w_i \neq 0\}$, in particular the support size $|\mathrm{supp}(w)|$;
- the *height* $\|w\|_\infty = \max_i |w_i|$.

We say $w$ is *sparse* if $|\mathrm{supp}(w)| \leq n/2$ (rounded up) and *full* otherwise. We record the height separately rather than folding it into a low/mid/high taxonomy. The two odd-rank orbits at $n = 9$ used in §7 arise from a sparse height-one representative

$$M \;:=\; (-1, -1, 0, 0, 0, 0, 0, 1, 1) \;\in\; A_8$$

and a sparse height-two representative

$$O_6 \;:=\; (-2, -1, 0, 0, 0, 0, 0, 1, 2) \;\in\; A_8.$$

Both have support size 4, but $\|M\|_\infty = 1$ versus $\|O_6\|_\infty = 2$; this height invariant is sufficient to separate the two $G^*$-orbits of §7.

### 6.3 Combinatorial identities from kernel vectors

Each kernel certificate $w \in A_{n-1}$ encodes a linear identity $\gamma(L)\, w = 0$, which expands as

$$\sum_{j=1}^{n} w_j\, \gamma(L_{ij}) \;=\; 0 \qquad \text{for every row } i \text{ of } L, \tag{23}$$

A left-kernel certificate gives the analogous column identities. For sparse right certificates, (23) reduces to a relation among at most $|\mathrm{supp}(w)|$ column entries of $\gamma(L)$ per row. The resulting identity is Sudoku-specific only insofar as the certificate $w$ arises from the Sudoku rank-drop data; it is not a generic consequence of the Latin-square axioms.

# 7 Orbit structure and $A_8$ saturation

We now restrict to $n = 9$ and analyze the $G^*$-orbits of the kernel certificates of §6.2. For every $n$ used below, we take

$$G_n^* := S_n \times \{\pm 1\}, \qquad (\sigma, \varepsilon) \cdot v := \varepsilon\,(v_{\sigma^{-1}(1)}, \ldots, v_{\sigma^{-1}(n)})$$

on $A_{n-1} \subset \mathbb{Z}^n$. Thus $G_n^*$ is the signed coordinate- permutation group generated by coordinate permutations and global sign. When $n = 9$ we write $G^*$ for $G_9^*$.

## 7.1 Sparse height-one and height-two orbits

Acting on $A_8$ by signed permutation of coordinates, $G^*$ produces two finite orbits of the representatives $M$ and $O_6$:

$$\operatorname{orb}(M) \subseteq A_8, \qquad |\operatorname{orb}(M)| = 756,$$

$$\operatorname{orb}(O_6) \subseteq A_8, \qquad |\operatorname{orb}(O_6)| = 3024.$$

The two orbits are disjoint (the height invariant $\|\cdot\|_\infty$ separates them) and each is a finite set of integer vectors of $V_{\mathrm{std},n}$.

## 7.2 Lattice generated by an orbit

For an orbit $\mathcal{O} \subseteq A_8$, write

$$\Lambda_{\mathcal{O}} := \mathbb{Z}\langle \mathcal{O} \rangle$$

for the integer lattice generated by $\mathcal{O}$ inside $V_{\mathrm{std},n}$. By construction $\Lambda_{\mathcal{O}} \subseteq A_8$. The *saturation* statement is the equality $\Lambda_{\mathcal{O}} = A_8$; equivalently, the elementary divisors of any matrix whose columns are vectors of $\mathcal{O}$ spanning $\Lambda_{\mathcal{O}}$ should all equal 1 (rank 8).

**Certified Proposition 7.1** ($A_8$ saturation)**.** *At $n = 9$,*

$$\Lambda_M = \Lambda_{O_6} = A_8 = V_{\mathrm{std},n} \cap \mathbb{Z}^9.$$

*Both $\Lambda_M$ and $\Lambda_{O_6}$ have rank 8 and elementary divisors* $(1, 1, 1, 1, 1, 1, 1, 1)$.

*Certification.* For each orbit $\mathcal{O} \in \{\operatorname{orb}(M), \operatorname{orb}(O_6)\}$, the entire orbit is used as a (redundant) spanning set: the orbit vectors are stacked into the rows of an integer matrix of shape $(|\mathcal{O}|, 9)$, namely $(756, 9)$ for $\operatorname{orb}(M)$ and $(3024, 9)$ for $\operatorname{orb}(O_6)$. Smith normal form over $\mathbb{Z}$ is applied via SymPy. In both cases the elementary divisors are $(1, 1, 1, 1, 1, 1, 1, 1)$ (eight 1's), confirming rank 8 and full saturation of $A_8$ inside $V_{\mathrm{std},n}$. The implementation is in `phase_9_18_S9q_beta_lattice.py`; the certified output is `A8_saturation.json`. □

## 7.3 The $\{M, O_6\}$ dichotomy at $n = 9$

Certified Proposition 7.1 promotes the height taxonomy of §6.2 to a structural statement: at $n = 9$, the sparse-template odd-rank certificates of $E_{\gamma,L}$ split into exactly two $G^*$-orbits, with representatives $M$ (sparse height-one, $|\mathrm{orb}| = 756$) and $O_6$ (sparse height-two, $|\mathrm{orb}| = 3024$), and both orbits saturate $A_8$. The general-$n$ extension of this dichotomy is the content of §9.

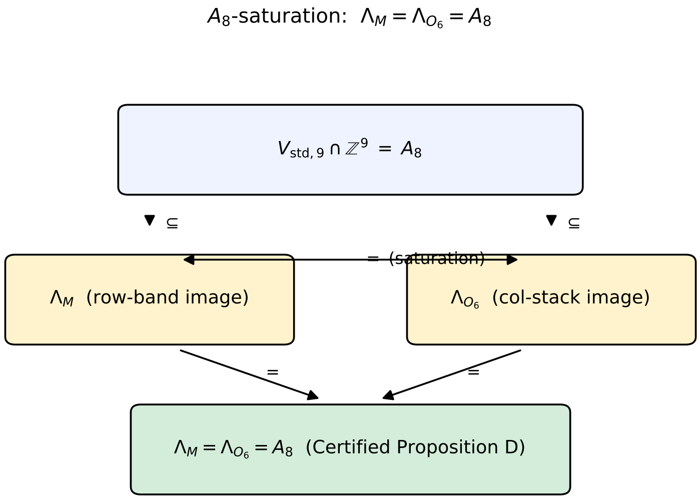


Figure 3: Schematic of the saturation $\Lambda_M = \Lambda_{O_6} = A_8$. Both sparse height-one and sparse height-two orbits already saturate the ambient root lattice $A_8$ at the level of $\mathbb{Z}$-spans.

# 8 The $M_{19}$ laboratory and affine coset symmetry

The Sudoku grid $M_{19}$ is the laboratory base under which the bilateral symmetry of $E_{\gamma,L}$ is most cleanly visible at $n = 9$. The grid itself is

$$M_{19} \;=\; \begin{pmatrix} 2 & 6 & 3 & 8 & 1 & 5 & 4 & 7 & 9 \\ 4 & 9 & 8 & 3 & 6 & 7 & 2 & 5 & 1 \\ 7 & 5 & 1 & 4 & 9 & 2 & 3 & 6 & 8 \\ 8 & 2 & 5 & 1 & 3 & 9 & 6 & 4 & 7 \\ 6 & 4 & 7 & 5 & 2 & 8 & 1 & 9 & 3 \\ 3 & 1 & 9 & 6 & 7 & 4 & 5 & 8 & 2 \\ 5 & 7 & 4 & 2 & 8 & 1 & 9 & 3 & 6 \\ 9 & 3 & 2 & 7 & 5 & 6 & 8 & 1 & 4 \\ 1 & 8 & 6 & 9 & 4 & 3 & 7 & 2 & 5 \end{pmatrix}. \tag{24}$$

This grid is recorded byte-equal in the certified package; it emerged from a seeded Markov-walk search over Sudoku bases, conducted as described in §11, as a candidate with the largest observed odd-rank relabeling count among the sampled grids. We make no claim that $M_{19}$ is globally maximal in this respect, nor that it is unique up to isomorphism among grids with 100 odd-rank relabelings; it is fixed here as the laboratory base on which the bilateral affine coset symmetry of §8.3 is most cleanly observed.

## 8.1 Exhaustive $S_9$ audit of $M_{19}$

We separate three distinct objects that all live on the laboratory base $M_{19}$ and should not be conflated into a single symbol $\Gamma^*_{M_{19}}$.

**Definition 8.1** (Three sets on $M_{19}$)**.** At the laboratory base $M_{19}$ we define:

(a) The exhaustive *rank locus*

$$\Gamma_{\mathrm{rank}}(M_{19}) \;:=\; \{\gamma \in S_9 : \mathrm{rank}\,\widehat{E}_{\gamma,M_{19}} = 7\},$$

the set of all relabelings that drop the rank of the centered operator below the generic value 8.

(b) The *shared left-kernel subset*

$$\Gamma^*_{M_{19}} := \Gamma_{\mathrm{rank}}(M_{19}) \cap \Gamma_L(M_{19}, w_9^L) = \{\gamma \in \Gamma_{\mathrm{rank}}(M_{19}) : \ker(E_{\gamma,L}^\top) \cap V_{\mathrm{std},n} = \mathbb{Q}\langle w_9^L\rangle\},$$

equivalently the relabelings satisfying $w_9^{L\top} E_{\gamma,M_{19}} = 0$, where $w_9^L$ is defined below.

(c) The *ambient coordinate stabilizer*

$$\mathrm{Stab}_{S_9}(w_9^L) \subset S_9,$$

the coordinate stabilizer of $w_9^L$ under the standard action of $S_9$ on $\mathbb{Z}^9$.

The three sets are pairwise distinct as cardinalities and as combinatorial objects: (a) is a set of permutations, (b) is a proper subset of (a), and (c) is a subgroup of $S_9$.

Throughout the sequel, $\Gamma^*_{M_{19}}$ always denotes the shared left-kernel subset in Definition 8.1(b), not the full rank-locus $\Gamma_{\mathrm{rank}}(M_{19})$ in Definition 8.1(a). Thus $|\Gamma^*_{M_{19}}| = 72$ and $|\Gamma_{\mathrm{rank}}(M_{19})| = 100$ are intentionally distinct certified quantities.

**Proposition 8.2** (Audit of $M_{19}$)**.** *The exhaustive $S_9$-audit of $M_{19}$ yields:*

(i) $|\Gamma_{\mathrm{rank}}(M_{19})| = 100$ *out of* $|S_9| = 9! = 362{,}880$ *(rank distribution* $\{8 : 362{,}780,\ 7 : 100\}$*);*

(ii) *the shared left-kernel subset has size* $|\Gamma^*_{M_{19}}| = 72$*, with shared primitive left-kernel direction*

$$w_9^L = (1,\,1,\,1,\,-2,\,-2,\,-2,\,1,\,1,\,1) \in A_8;$$

*the* $|\Gamma_{\mathrm{rank}}(M_{19}) \setminus \Gamma^*_{M_{19}}| = 28$ *residual relabelings are rank-7 relabelings but do not satisfy this shared left-kernel condition;*

(iii) *the ambient coordinate stabilizer satisfies*

$$\mathrm{Stab}_{S_9}(w_9^L) \cong S_3 \times S_6, \qquad |\mathrm{Stab}_{S_9}(w_9^L)| = 3! \cdot 6! = 4320,$$

*since $w_9^L$ has value-multiplicities* $\{-2 : 3,\ 1 : 6\}$.

*Certification.* The rank distribution $\{8 : 362{,}780,\ 7 : 100\}$ is recomputed by an exhaustive two-stage scan: (a) compute the determinant of the projected $8 \times 8$ matrix $\widehat{E}_{\gamma,M_{19}}$ modulo the prime 1,000,003 for every $\gamma \in S_9$ and flag the 100 zero candidates; because a zero rational determinant is zero modulo every prime, this modular filter cannot miss a rank-drop relabeling; (b) apply Sympy exact integer rank verification to every candidate, returning rank 7 for all 100. The full list of 100 permutations is shipped with a SHA-256 manifest in the certified package. Statement (ii) is checked twice: first in the original rank-locus audit, and then in the middle-band certificate. The same 72 relabelings are exactly those for which $w_9^{L\top} E_{\gamma,M_{19}} = 0$, and this 72-element subset is checked directly to lie inside the 100 rank-7 relabelings. Statement (iii) is the orbit-stabilizer calculation: $w_9^L$ has the multiset $\{1, 1, 1, -2, -2, -2, 1, 1, 1\}$ with multiplicities $\{-2 : 3,\ 1 : 6\}$, hence its $S_9$-coordinate stabilizer is $S_3 \times S_6$ of order 4320. The orbit of $w_9^L$ in $S_9 \cdot w_9^L$ has size $9!/4320 = 84$; this ambient orbit is not the rank locus and is not used to count $\Gamma^*_{M_{19}}$. □

*Remark* 8.3 (The number 4320 is an ambient stabilizer, not an orbit). The number 4320 is the order of the ambient coordinate stabilizer $\mathrm{Stab}_{S_9}(w_9^L) = \mathrm{Stab}_{S_9}(w_9^L) \cong S_3 \times S_6$, an algebraic invariant of the shared left-kernel direction $w_9^L$. It is *not* the cardinality of any of $\Gamma_{\mathrm{rank}}(M_{19})$, $\Gamma^*_{M_{19}}$, or any orbit of $K_L \times K_R$ on a rank-locus set; indeed $|K_L \times K_R| = 72 \cdot 72 = 5184$ is not divisible by 4320 (ratio 6/5). The three numbers $|\Gamma_{\mathrm{rank}}(M_{19})| = 100$, $|\Gamma^*_{M_{19}}| = 72$, $|\mathrm{Stab}_{S_9}(w_9^L)| = 4320$ live in distinct categories (rank-locus size, shared left-kernel subset size, ambient stabilizer order) and should not be identified.

### 8.2 Single-coset structure of the shared left-kernel subset $\Gamma^*_{M_{19}}$

We now identify the symmetries of the shared left-kernel subset $\Gamma^*_{M_{19}}$. Define

$$K_L := \{\pi \in S_9 : \pi \cdot \Gamma^*_{M_{19}} \subseteq \Gamma^*_{M_{19}}\}, \tag{25}$$
$$K_R := \{\pi \in S_9 : \Gamma^*_{M_{19}} \cdot \pi \subseteq \Gamma^*_{M_{19}}\}. \tag{26}$$

By a direct exhaustive computation in $S_9$, recorded in the M19 audit certificate, both $K_L$ and $K_R$ are subgroups of $S_9$ of order 72, and $\Gamma^*_{M_{19}}$ is simultaneously a single left coset of $K_L$ and a single right coset of $K_R$.

**Certified Proposition 8.4** (Single-coset structure of $\Gamma^*_{M_{19}}$)**.** *The shared left-kernel subset $\Gamma^*_{M_{19}}$ has cardinality 72, and there exists $\pi_0 \in \Gamma^*_{M_{19}}$ such that*

$$\Gamma^*_{M_{19}} = K_L \cdot \pi_0 = \pi_0 \cdot K_R, \tag{27}$$

*with $|K_L| = |K_R| = 72$. The two subgroups intersect trivially, $K_L \cap K_R = \{e\}$, but they are conjugate in $S_9$ via $\pi_0$:*

$$K_L = \pi_0 \, K_R \, \pi_0^{-1}. \tag{28}$$

*Equivalently, the double coset $K_L \cdot \pi_0 \cdot K_R$ has cardinality 72, which by the standard double-coset formula*

$$|K_L \, \pi_0 \, K_R| = \frac{|K_L| \cdot |K_R|}{|K_L \cap \pi_0 \, K_R \, \pi_0^{-1}|}$$

*forces $|K_L \cap \pi_0 \, K_R \, \pi_0^{-1}| = 72 = |K_L| = |K_R|$, i.e. (28).*

*Certification.* The orders $|K_L| = |K_R| = 72$ and the equality $\Gamma^*_{M_{19}} = K_L \cdot \pi_0 = \pi_0 \cdot K_R$ are read off the exhaustive set-stabilizer computation recorded in the M19 audit certificate. The intersection $K_L \cap K_R$ is computed as a finite set in $S_9$ and returns the singleton $\{e\}$. The conjugacy $K_L = \pi_0 K_R \pi_0^{-1}$ is then forced by the double-coset size identity: with $|K_L \, \pi_0 \, K_R| = |\Gamma^*_{M_{19}}| = 72$ and $|K_L| = |K_R| = 72$, one gets $|K_L \cap \pi_0 K_R \pi_0^{-1}| = 72$, hence $K_L = \pi_0 K_R \pi_0^{-1}$; the affine certificate records the same conjugacy check directly. □

A schematic of the single-coset structure is shown in Figure 4.

### 8.3 Affine identification: $K_L \cong K_R \cong 3^2{:}D_4$

**Certified Proposition 8.5** (Affine coset symmetry of $M_{19}$)**.** *At $L = M_{19}$ and $n = 9$, the left and right affine coset-symmetry groups satisfy*

$$K_L \cong K_R \cong 3^2{:}D_4.$$

*Equivalently, each has a normal regular translation subgroup $T \cong (\mathbb{Z}/3)^2$ and a point stabilizer $H \cong D_4$, so $K_L \cong T \rtimes H$ and the same holds for $K_R$. Both groups have order 72, exponent 12, derived series of orders $(72, 18, 9, 1)$, abelianization $(\mathbb{Z}/2)^2$, trivial centre, and 9 conjugacy classes. The element-order distribution is $\{1{:}1,\ 2{:}21,\ 3{:}8,\ 4{:}18,\ 6{:}24\}$. $K_L$ and $K_R$ are not equal as subgroups of $S_9$ (their intersection is trivial: $K_L \cap K_R = \{e\}$), but they are conjugate inside $S_9$ via an external structural permutation $\pi_0 \in S_9 \setminus (K_L \cup K_R)$ characterized in Remark 8.6. Neither $K_L$ nor $K_R$ is contained in the ambient coordinate stabilizer $\mathrm{Stab}_{S_9}(w_9^L)$.*

*Certification of the group identification.* We describe $K_L$; the computation for $K_R$ is identical. Let $T$ be the identity together with the eight elements of $K_L$ of order 3. The certificate verifies that $T$ is closed, abelian, normal in $K_L$, and regular on the nine symbols. Its element-order

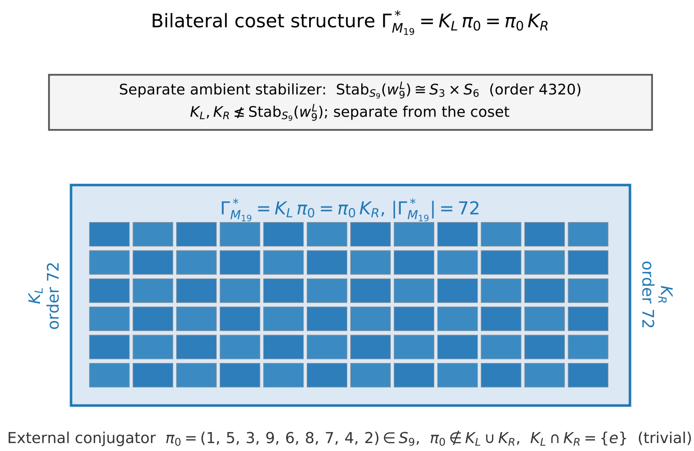


Figure 4: Schematic of the single-coset structure of $\Gamma^*_{M_{19}}$. The left axis is the $|K_L| = 72$ direction; $\Gamma^*_{M_{19}} = K_L \cdot \pi_0$ is itself a single left $K_L$-coset of cardinality 72. The structural permutation $\pi_0 \in \Gamma^*_{M_{19}}$ also satisfies $\Gamma^*_{M_{19}} = \pi_0 \cdot K_R$; the conjugation $K_L = \pi_0\, K_R\, \pi_0^{-1}$ identifies the two affine sides.

distribution is $\{1{:}1, 3{:}8\}$, which excludes $\mathbb{Z}/9$ and hence identifies $T \cong (\mathbb{Z}/3)^2$. Let $H \leq K_L$ be the stabilizer of the point 1. Then $|H| = 8$, and its element-order distribution is

$$\{1{:}1, 2{:}5, 4{:}2\},$$

which uniquely identifies $H$ with $D_4$ among groups of order 8. Since $T$ is normal, $T \cap H = 1$, and $|T|\,|H| = 9{\cdot}8 = 72 = |K_L|$, we obtain $K_L = T \rtimes H \cong 3^2{:}D_4$. The quotient $K_L/T$ has the same $D_4$ order fingerprint. The derived subgroup has order 18, and the quotient by the derived subgroup has order distribution $\{1{:}1, 2{:}3\}$, giving abelianization $(\mathbb{Z}/2)^2$. The full identification certificate, including the 8 explicit generator permutations and the ambient-stabilizer non-containment check, is recorded in `M19_KL_KR_affine_D4.json`. □

*Remark* 8.6 (Structural permutation $\pi_0$). The canonical structural permutation conjugating $K_R$ to $K_L$ is the lex-minimum element of the bilateral coset $\Gamma^*_{M_{19}} = K_L\pi_0 = \pi_0 K_R$, namely

$$\pi_0 \;=\; (1,\,5,\,3,\,9,\,6,\,8,\,7,\,4,\,2) \;\in\; S_9 \qquad \text{(one-line notation)}. \tag{29}$$

This $\pi_0$ is *not* an element of $K_L$ nor of $K_R$; in particular it is not one of the eight generators of $K_L$. It satisfies $\pi_0 K_R \pi_0^{-1} = K_L$ and the entire coset $\Gamma^*_{M_{19}}$ admits the bilateral factorisation $\Gamma^*_{M_{19}} = \pi_0 K_R = K_L \pi_0$. The affine certificate records both the eight-generator presentation of $K_L$ and the explicit value of $\pi_0$.

## 8.4 Middle-band hypergraph realization

The middle-band balance criterion also has a finite hypergraph formulation. Let $\mathcal{H}^{\mathrm{mid}}_{M_{19}}$ be the reduced 3-uniform hypergraph on the symbol set $\{1, \ldots, 9\}$ whose edges are the distinct unordered triples appearing in rows 4, 5, 6 of a single column of $M_{19}$. Explicitly,

$$\mathcal{H}^{\mathrm{mid}}_{M_{19}} = \{\{1,2,4\},\ \{1,5,6\},\ \{2,3,7\},\ \{3,6,8\},\ \{4,8,9\},\ \{5,7,9\}\}.$$

The nine columns give these six edges with multiplicities $(1, 2, 2, 1, 2, 1)$ in the displayed order; the reduced hypergraph forgets those multiplicities. Let

$$\mathcal{T}_{15} := \{\{a, b, c\} \subset \{1, \ldots, 9\} : a + b + c = 15\}$$

be the 8-edge magic-triple hypergraph, and let

$$\mathcal{T}_{15}^{-} = \{\{1,5,9\},\ \{1,6,8\},\ \{2,4,9\},\ \{2,6,7\},\ \{3,4,8\},\ \{3,5,7\}\}$$

be the 6-edge subhypergraph obtained from $\mathcal{T}_{15}$ by removing $\{2,5,8\}$ and $\{4,5,6\}$.

**Certified Proposition 8.7** (M19 middle-band hypergraph realization)**.** *With $\pi_0$ as in Remark 8.6,*

$$\Gamma^*_{M_{19}} = \{\gamma \in S_9 : \gamma(e) \in \mathcal{T}_{15} \text{ for every } e \in \mathcal{H}^{\mathrm{mid}}_{M_{19}}\}. \tag{30}$$

*Moreover every relabeling in this set has the same reduced image $\mathcal{T}_{15}^{-}$, and*

$$\mathrm{Aut}(\mathcal{H}^{\mathrm{mid}}_{M_{19}}) = K_R, \qquad \mathrm{Aut}(\mathcal{T}_{15}^{-}) = K_L. \tag{31}$$

*Consequently*

$$\Gamma^*_{M_{19}} = \pi_0\ \mathrm{Aut}(\mathcal{H}^{\mathrm{mid}}_{M_{19}}) = \mathrm{Aut}(\mathcal{T}_{15}^{-})\,\pi_0. \tag{32}$$

*Both reduced automorphism groups have order* 72*. In contrast, the full magic-triple hypergraph $\mathcal{T}_{15}$ has automorphism group of order* 8*.*

*Certification.* For a relabeling $\gamma$, the condition $\gamma(e) \in \mathcal{T}_{15}$ for every middle-band edge $e$ says exactly that each middle-band column triple $(\gamma(M_{4j}), \gamma(M_{5j}), \gamma(M_{6j}))$ has sum 15. By Proposition 8.2, this is equivalent to $\gamma \in \Gamma^*_{M_{19}}$ and to the left-kernel equation $w_9^{L\top} E_{\gamma,M_{19}} = 0$. The certificate `M19_middle_band_balance.json` exhaustively scans all 9! relabelings and records the equality of the embedding set with $\Gamma^*_{M_{19}}$.

The same certificate computes the reduced source and target automorphism groups as finite subgroups of $S_9$. It checks $|\,\mathrm{Aut}(\mathcal{H}^{\mathrm{mid}}_{M_{19}})| = 72$ and identifies this group exactly with $K_R$; it also checks $|\,\mathrm{Aut}(\mathcal{T}_{15}^{-})| = 72$ and identifies this group exactly with $K_L$. The permutation $\pi_0$ maps $\mathcal{H}^{\mathrm{mid}}_{M_{19}}$ onto $\mathcal{T}_{15}^{-}$, so the set of all such isomorphisms is both $\pi_0\,\mathrm{Aut}(\mathcal{H}^{\mathrm{mid}}_{M_{19}})$ and $\mathrm{Aut}(\mathcal{T}_{15}^{-})\pi_0$, giving (32). Finally, the verifier separately computes $|\,\mathrm{Aut}(\mathcal{T}_{15})| = 8$, so the 72-element phenomenon is not an automorphism group of the full magic-triple hypergraph; it is an isomorphism torsor between the M19 middle-band hypergraph and the six-edge magic subhypergraph $\mathcal{T}_{15}^{-}$. □

*Remark* 8.8 (No Hessian or SmallGroup claim)*.* The certificate for Certified Proposition 8.5 does not assert a `SmallGroup` identifier and does not identify $K_L$ or $K_R$ with the classical Hessian group $3^2{:}Q_8$. The point-stabilizer fingerprint $\{1:1, 2:5, 4:2\}$ separates $D_4$ from $Q_8$, whose order fingerprint is $\{1:1, 2:1, 4:6\}$. The Sudoku specificity of the affine $3^2{:}D_4$ realization is investigated in §10.

# 9 Stabilizer dichotomy $\{M_n, O_n\}$

We now lift the $\{M, O_6\}$ structure of §7.3 to a one-parameter family of representatives at general $n$. For $n \geq 5$, define the canonical sparse height-one and height-two integer vectors of $A_{n-1}$ by

$$M_n \;:=\; (-1,\,-1,\,\underbrace{0,\ldots,0}_{n-4},\,1,\,1), \qquad O_n \;:=\; (-2,\,-1,\,\underbrace{0,\ldots,0}_{n-4},\,1,\,2).$$

At $n = 9$, $M_9 = M$ and $O_9 = O_6$ in the notation of §6.2. The first two results record the finite certified evidence base; the third gives the general proof of the stabilizer and lattice-saturation dichotomy for all $n \geq 5$.

## 9.1 Exhaustive verification at $5 \le n \le 13$

**Certified Proposition 9.1** (Exhaustive dichotomy)**.** *For every integer $n$ with $5 \le n \le 13$,*

$$|\operatorname{Stab}_{G_n^*}(M_n)| \;=\; 8\,(n-4)!, \qquad |\operatorname{Stab}_{G_n^*}(O_n)| \;=\; 2\,(n-4)!,$$

*and*

$$\begin{aligned} |\operatorname{orb}_{G_n^*}(M_n)| \;&=\; \frac{n(n-1)(n-2)(n-3)}{4}, \\ |\operatorname{orb}_{G_n^*}(O_n)| \;&=\; n(n-1)(n-2)(n-3). \end{aligned}$$

*The orbit-distance $d_{M_n}(\operatorname{orb}(O_n)) = 2$, and $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$ (saturation, elementary divisors all $1$).*

Here $d_{M_n}(\operatorname{orb}(O_n))$ denotes the minimum Hamming distance from $M_n$ to the signed-coordinate orbit of $O_n$, after identifying vectors by their coordinate tuples. The certified proposition is verified by direct computation in `phase_9_18_S9q_gamma1.py`; total wall-time $\approx 31$ seconds. The certified output is `stabilizers_5_to_13.json`.

## 9.2 Analytic identification at $n \in \{7, 9, 11\}$

**Certified Proposition 9.2** (Certified dichotomy at $n \in \{7, 9, 11\}$)**.** *For each $n \in \{7, 9, 11\}$,*

$$\operatorname{Stab}_{G_n^*}(M_n) \;\cong\; D_4 \times S_{n-4}, \qquad \operatorname{Stab}_{G_n^*}(O_n) \;\cong\; \mathbb{Z}/2 \times S_{n-4},$$

*and the integer hulls satisfy $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$.*

*Certification and analytic reduction.* At each $n \in \{7, 9, 11\}$, the stabilizer of $M_n$ in $G_n^*$ admits a canonical short exact sequence

$$1 \;\to\; \ker\psi \;\to\; \operatorname{Stab}_{G_n^*}(M_n) \;\xrightarrow{\psi}\; S_{n-4} \;\to\; 1,$$

where $\psi$ is the restriction to the zero-coordinate block of $M_n$ and $\ker\psi$ acts only on the four nonzero coordinates. The sequence splits because the inclusion $S_{n-4} \hookrightarrow G_n^*$ on the zero block is a section. Hence

$$\operatorname{Stab}_{G_n^*}(M_n) \;\cong\; \ker\psi \rtimes_{\mathrm{triv}} S_{n-4} \;=\; \ker\psi \,\times\, S_{n-4},$$

and the order of $\ker\psi$ is $8 = |\operatorname{Stab}_{G_n^*}(M_n)|/(n-4)!$ by Proposition 9.1.

To identify $\ker\psi$ as $D_4$ rather than as one of the four other groups of order 8, we compute the order distribution of $\ker\psi$ and obtain

$$\{1{:}1,\;\; 2{:}5,\;\; 4{:}2\},$$

which uniquely separates $D_4$ from $Q_8$ ($\{1{:}1, 2{:}1, 4{:}6\}$), $\mathbb{Z}/8$ ($\{1{:}1, 2{:}1, 4{:}2, 8{:}4\}$), $\mathbb{Z}/4 \times \mathbb{Z}/2$ ($\{1{:}1, 2{:}3, 4{:}4\}$), and $(\mathbb{Z}/2)^3$ ($\{1{:}1, 2{:}7\}$).

The argument for $\operatorname{Stab}_{G_n^*}(O_n) \cong \mathbb{Z}/2 \times S_{n-4}$ is identical, with $\ker\psi$ of order 2 and order distribution $\{1{:}1,\; 2{:}1\}$ (the unique group of order 2).

The lattice saturation $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$ is read from the elementary divisors of the orbit-Gram matrix, all equal to 1 at $n \in \{7, 9, 11\}$. $\square$

The full witness fingerprint, including the order-multiset checks and the alternative-exclusion list, is in `dichotomy_n7_n9_n11.json`.

### 9.3 General-$n$ theorem

**Theorem 9.3** (General-$n$ dichotomy)**.** *For every integer $n \geq 5$,*

$$\mathrm{Stab}_{G_n^*}(M_n) \;\cong\; D_4 \times S_{n-4}, \qquad \mathrm{Stab}_{G_n^*}(O_n) \;\cong\; \mathbb{Z}/2 \times S_{n-4},$$

*and $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$.*

*Proof.* Recall that $G_n^* = S_n \times \{\pm 1\}$ acts by coordinate permutations and global sign. For $M_n$, the coordinates split into two entries equal to $-1$, two entries equal to 1, and an $(n-4)$-element zero block. The zero block may be permuted freely, giving an $S_{n-4}$ factor. On the four nonzero coordinates, the sign-preserving part permutes the two negative entries and the two positive entries independently, while multiplication by $-1$ swaps the negative pair with the positive pair. Thus the nonzero-coordinate factor is $(S_2 \times S_2) \rtimes C_2 \cong D_4$, and it commutes with the zero-block factor. Hence $\mathrm{Stab}_{G_n^*}(M_n) \cong D_4 \times S_{n-4}$.

For $O_n$, the nonzero entries $-2, -1, 1, 2$ are all distinct. The zero block again gives $S_{n-4}$, and the only nontrivial operation on the nonzero coordinates is global sign, which exchanges $-2 \leftrightarrow 2$ and $-1 \leftrightarrow 1$. Therefore $\mathrm{Stab}_{G_n^*}(O_n) \cong C_2 \times S_{n-4}$.

It remains to prove saturation. Each orbit vector has coordinate sum zero, so both orbit lattices are contained in $A_{n-1}$. Conversely, fix distinct indices $i$ and $k$. Since $n \geq 5$, choose three further indices $j, a, b$ distinct from $i, k$. In the $G_n^*$-orbit of $M_n$, compare the two vectors with nonzero support arranged as

$$-1 \text{ at } i, j, \quad 1 \text{ at } a, b,$$

and

$$-1 \text{ at } k, j, \quad 1 \text{ at } a, b.$$

Their difference is $e_k - e_i$. Thus every root $e_k - e_i$ lies in $\Lambda_{M_n}$, so $\Lambda_{M_n} = A_{n-1}$. For $O_n$, keep the positions of $-2$, 1, and 2 fixed at three indices distinct from $i, k$, and move only the coordinate carrying $-1$ from $i$ to $k$. The difference of the two orbit vectors is again $e_k - e_i$. Hence $\Lambda_{O_n} = A_{n-1}$ as well. □

Certified Proposition 9.1 remains useful as an independent finite audit: in addition to the formulas now proved in Theorem 9.3, it checks the orbit-distance statement $d_{M_n}(\mathrm{orb}(O_n)) = 2$ for $5 \leq n \leq 13$.

## 10 Sudoku-specificity beyond Latin-square autotopisms

The affine coset-symmetry group $K_L \cong 3^2{:}D_4$ identified at $M_{19}$ is not a familiar Latin-square invariant. The classical group attached to a Latin square $L$ is its autotopism group $\mathrm{Aut}(L)$, i.e. the set of triples $(\alpha, \beta, \gamma) \in S_n^3$ that preserve $L$ in the sense $L_{\alpha(i)\,\beta(j)} = \gamma(L_{ij})$. Even when $L$ is taken as a Sudoku grid, the operative group from the Latin-square literature is $\mathrm{Aut}(L)$ or one of its standard refinements (paratopism, principal isotopy, see [17, 16, 6]). This section shows that for $L = L_{M_{19}}$ the autotopism group is trivial, hence cannot be the source of $K_L$.

### 10.1 Triviality of $\mathrm{Aut}(L_{M_{19}})$

Let $L_{M_{19}}$ denote the Latin square underlying $M_{19}$; that is, the $9 \times 9$ array of (24) viewed without its Sudoku/box structure.

**Certified Proposition 10.1** ($\mathrm{Aut}(L_{M_{19}})$ is trivial)**.** *The autotopism group of $L_{M_{19}}$ has order 1:*

$$|\,\mathrm{Aut}(L_{M_{19}})| \;=\; 1.$$

*Certification.* The full $9!^3 = 4.79 \times 10^{16}$ triple-search is reduced algorithmically: fix $\gamma \in S_9$, and for each fixed $\gamma$ search the $9!^2$ pairs $(\alpha, \beta)$ such that $L_{\alpha(i),\beta(j)} = \gamma(L_{ij})$ via a constrained backtracking that prunes on the first row. The exhaustive scan over all $9! = 362{,}880$ values of $\gamma$ yields the trivial autotopism $(\mathrm{id}, \mathrm{id}, \mathrm{id})$ and no other. The certificate `Aut_LM19_trivial.json` records the deterministic run metadata. □

**Corollary 10.2** ($K_L$ is not an autotopism group)**.** *The affine coset-symmetry group $K_L$ of $M_{19}$ is not the image of any autotopism subgroup of $L_{M_{19}}$. In particular, $|K_L| = 72 \neq 1 = |\operatorname{Aut}(L_{M_{19}})|$.*

*Proof.* Any homomorphic image of $\operatorname{Aut}(L_{M_{19}})$ in $S_9$ has order $\leq |\operatorname{Aut}(L_{M_{19}})| = 1$ by Certified Proposition 10.1. But $|K_L| = 72$ by Certified Proposition 8.5, so $K_L$ cannot be such an image. □

## 10.2 Middle-band balance and the $A_8$ mechanism for $K_L$

Since $K_L$ is not generated by autotopisms of $L_{M_{19}}$, the Latin-square structure alone does not explain this 72-element affine symmetry. In this grid it is detected by the middle row-band balance condition.

**Lemma 10.3** (Middle-band balance criterion)**.** *Let $L = M_{19}$ and let*

$$w = (1, 1, 1, -2, -2, -2, 1, 1, 1) \in A_8.$$

*For $\gamma \in S_9$, the left-kernel condition*

$$w^\top E_{\gamma,L} = 0$$

*is equivalent to the column-wise middle-band balance condition*

$$\sum_{i=4}^{6} \gamma(L_{ij}) = 15 \qquad (1 \leq j \leq 9). \tag{33}$$

*Proof.* Fix a column $j$ and put

$$B_j := \sum_{i=4}^{6} \gamma(L_{ij}).$$

Because $L$ is a Latin square, the $j$-th column of $\gamma(L)$ is a permutation of $\{1, \dots, 9\}$ and therefore has total sum 45. Since $\sum_i w_i = 0$, centering does not change the weighted column sum:

$$\left(w^\top E_{\gamma,L}\right)_j = \sum_{i=1}^{9} w_i \, \gamma(L_{ij}).$$

The entries outside rows $4, 5, 6$ have total sum $45 - B_j$, while the middle-band entries have coefficient $-2$. Hence

$$\left(w^\top E_{\gamma,L}\right)_j = (45 - B_j) - 2B_j = 45 - 3B_j.$$

Thus the $j$-th component vanishes if and only if $B_j = 15$, and the claim follows column by column. □

**Proposition 10.4** (Exact middle-band realization of $\Gamma^*_{M_{19}}$)**.** *For $L = M_{19}$, the shared left-kernel subset is exactly*

$$\Gamma^*_{M_{19}} = \left\{ \gamma \in S_9 : \sum_{i=4}^{6} \gamma(L_{ij}) = 15 \text{ for every } j = 1, \dots, 9 \right\}.$$

*This set has cardinality* 72.

*Certificate.* The analytic equivalence between (33) and $w^\top E_{\gamma,L} = 0$ is Lemma 10.3. The certificate `M19_middle_band_balance.json` exhaustively scans all $9! = 362{,}880$ relabelings in $S_9$ and records

$$\texttt{balanced\_count} = 72, \qquad \texttt{left\_kernel\_count} = 72, \qquad \texttt{Gamma\_star\_size} = 72,$$

and the two equality flags are true:

$$\texttt{balanced\_equals\_left\_kernel},$$
$$\texttt{balanced\_equals\_Gamma\_star}.$$

The same certificate lists the eight three-symbol subsets of $\{1, \ldots, 9\}$ with sum 15:

$$\{1, 5, 9\}, \{1, 6, 8\}, \{2, 4, 9\}, \{2, 5, 8\}, \{2, 6, 7\}, \{3, 4, 8\}, \{3, 5, 7\}, \{4, 5, 6\}.$$

□

Proposition 10.4, Certified Proposition 7.1, and Certified Proposition 8.5 give the structural chain used here: middle-band Sudoku balance cuts out $\Gamma^*_{M_{19}}$; the corresponding left-kernel certificate is a structured vector in $A_8$; and the left/right coset stabilizers of this class are affine groups $3^2{:}D_4$ on the nine symbols. The certified claim is this affine $D_4$ realization, not the classical Hessian group $3^2{:}Q_8$.

## 11 Computational methods and reproducibility

The four main theorems and the supporting propositions of this paper are obtained from a combination of analytic argument and exhaustive exact-arithmetic computation. This section records the methodology and points to the certified reproducibility package.

**Exact arithmetic.** All ranks, kernels, Smith normal forms, and group-order computations are performed in exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$, never in floating point. The integer rank computations use Bareiss fraction-free elimination [3]; rational nullspaces use SymPy RREF; Smith normal forms use the standard $\mathbb{Z}$-elementary-divisor algorithm. Each odd-rank report is independently cross-checked at primes $p \in \{2, 3, 5, 7\}$, the agreement of which rules out single-prime artefacts.

**Tier discipline.** Every numerical or structural claim in this paper is labeled with one of five tiers — THEOREM, EXHAUSTIVE, EMPIRICAL, OBSERVED TEMPLATE, CONJECTURE — following the convention introduced in [1, §8.26]. The composite tier THEOREM(EXHAUSTIVE) denotes an analytic theorem whose hypotheses include a finite exhaustive check (e.g., Certified Proposition 10.1). The complete tier ledger is in `certified/TIER_LEDGER.md`.

**Sampling caveats.** Wherever a corpus statistic appears in this paper (e.g., Remark 2.2 on 400 sampled grids), it is phrased as a finite-corpus proposition, not a population claim. The sampler used in such cases is a Markov-walk perturbation of a fixed Sudoku base, frozen at `BASE_SEED=20260440`; it does not produce a uniform distribution over Sud(9), and no universal-distribution claim is made.

**Reproducibility package.** Each result is paired with a certified JSON in `certified/` containing the inputs, outputs, source-data hashes, producer-script hashes, seed, UTC timestamp, paper label, and tier. The twelve JSONs are pinned by `MANIFEST.sha256`. The verifier `verify_all.py` re-checks the manifest and the claim-level invariants of each JSON; the regenerator `reproduce_all.py` rebuilds every certificate from the pinned source JSONs and the certificate builders, byte-equal to the manifest modulo the `produced_utc` timestamp. The upstream producer scripts are included and hash-checked as provenance; they are not rerun by default.

**Data and code availability.** The public companion repository is

https://github.com/aconsciousfractal/
Odd-Rank-Loci-and-Box-Induced-Symmetries-of-Centered-Sudoku-Operators.

The release artifact separates three levels of reproducibility: `verify_all.py` checks the manifest, schema, semantic invariants, and standalone provenance; `reproduce_all.py` rebuilds the twelve certificate JSONs byte-for-byte from the pinned source JSONs and builders; and the archived producer scripts in `scripts/legacy/` record the exact exhaustive scans from which those source JSONs were obtained. The last level is included and hash-checked for provenance, but is not part of the default replay because some scans are intentionally more expensive than the certificate rebuild.

# 12 Discussion and open problems

## 12.1 Summary

We have analyzed the centered Sudoku operator $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2} J_n$ on the standard hyperplane $V_{\text{std},n}$. The Box Band Lemma (Theorem 3.1) is an exact linear identity that is Sudoku-specific. The Lift Lemma (Theorem 6.1) places one-dimensional rational kernel lines in the natural primitive-integral certificate framework of the root lattice $A_{n-1}$. At $n = 9$ the sparse-template orbits $M$ and $O_6$ saturate $A_8$ (Certified Proposition 7.1), and on the laboratory grid $M_{19}$ the bilateral affine coset-symmetry group is $3^2{:}D_4$ (Certified Proposition 8.5); this symmetry is genuinely not explained by Latin-square autotopisms, since the underlying Latin square is rigid (Certified Proposition 10.1).

## 12.2 Open problems

Several questions remain open and are explicitly out of the present paper's scope.

(O1) **Cokernel structure of $E_{\gamma,L}$.** The integer Smith normal form of $E_{\gamma,L}$ at the 22 odd-rank relabelings of `base1` produces five distinct diagonals ((20)); only 10/22 are torsion-free, and the remaining torsion classes include $\mathbb{Z}/2$, $\mathbb{Z}/4$, $\mathbb{Z}/3$, and $\mathbb{Z}/2 \oplus \mathbb{Z}/10$. A closed-form classification of which $\gamma \in \Gamma^*_{\texttt{base1}}$ produces which torsion class is open, as is the analogous distribution for $M_{19}$'s 100 relabelings.

(O2) **Larger affine coset symmetries.** Does an analogue of the certified $3^2{:}D_4$ affine coset symmetry occur for Sudoku grids of larger square order, for example $n = 25$? The present paper verifies only the $n = 9$ instance at the laboratory grid $M_{19}$.

(O3) **Canonical-rank distribution at large $n$.** The sample histogram of rank $E_{\text{id},L}$ at $n = 9$ is $\{8{:}400\}$ on 400 Markov-walk samples. The high-corank tail $\{4, 6, 8\}$ predicted at small $n$ by the rank-parity draft is currently unobserved at $n = 9$ in any sample of moderate size.

(O4) **Trace–cokernel bridge.** A direct algebraic relation between $\operatorname{tr}(\widehat{E}_{\gamma,L}\, \widehat{E}^{\top}_{\gamma,L})$ and the cokernel structure is not known.

(O5) **General orbit-distance formula.** Certified Proposition 9.1 verifies $d_{M_n}(\operatorname{orb}(O_n)) = 2$ for $5 \le n \le 13$. A conceptual all-$n$ proof of this orbit-distance statement is not included here.

## 12.3 Out-of-scope, briefly

Two pointers, included as future-work signals only and *not* contributions of this paper.

The framework of §2 extends *verbatim* to several constrained Latin-square families: gerechte designs (general partition into equal-area boxes), rectangular Sudoku, orthogonal Sudoku Latin squares, and constraint-satisfaction problems with local symmetry groups. None of these are studied here.

Classical Hesse/Hessian groups, projective models of root lattices, and 4-dimensional polytope constructions are not used in the present arguments. In particular, this paper does not identify the certified $3^2{:}D_4$ group with the classical Hessian group $3^2{:}Q_8$.

# A Exhaustive base$_1$ odd-rank list

The full list of the 22 odd-rank symbol relabelings $\gamma \in S_9$ on `base1` is contained in the field `outputs.odd_perms_full_list` of `base1_odd_rank_22.json`. Each entry is a permutation of $\{1, \dots, 9\}$ in one-line notation. By construction, the cardinality 22 matches Certified Proposition 4.1; the four-way exact certification (rank, kernel, integer SNF, $\mathbb{F}_2$ SNF) is recorded in the same JSON under the `outputs.four_way_certificates` block. The certificate hash is pinned by `MANIFEST.sha256`.

# B $M_{19}$: generators of $K_L$ and affine $3^2{:}D_4$ data

The affine coset-symmetry group $K_L$ at $M_{19}$ is generated by the eight permutations of $\{1, \dots, 9\}$ recorded in the `outputs.K_L_generators_perms` field of `M19_KL_KR_affine_D4.json`. Their derived series of orders is $(72, 18, 9, 1)$, abelianization is $(\mathbb{Z}/2)^2$, exponent is 12, and the element-order multiset is $\{1:1,\ 2:21,\ 3:8,\ 4:18,\ 6:24\}$. The certificate identifies a normal regular translation subgroup $3^2$ and a point stabilizer $D_4$, hence $K_L \cong 3^2{:}D_4$. The same data hold for $K_R$, which is conjugate to $K_L$ via the structural permutation $\pi_0$ (Remark 8.6).

# C Verified-range stabilizer table for $n \in \{5, \dots, 13\}$

For each $n$ in $\{5, 6, 7, 8, 9, 10, 11, 12, 13\}$ the predicted and verified values of $|\operatorname{Stab}(M_n)|$, $|\operatorname{Stab}(O_n)|$, $|\operatorname{orb}(M_n)|$, $|\operatorname{orb}(O_n)|$, and the integer-hull elementary divisors of $\Lambda_{M_n}$ and $\Lambda_{O_n}$ are recorded in the `outputs.results` field of `stabilizers_5_to_13.json`. All nine rows pass `all_pass = true`. The total wall-time of the exhaustive verification is $\approx 31$ s.

# D Reproducibility manifest

The reproducibility package consists of:

- the twelve certified JSONs listed in §3 of `TIER_LEDGER.md`;
- the SHA-256 manifest `MANIFEST.sha256`;
- the verifier `verify_all.py` (claim-level invariant checks);
- the regenerator `reproduce_all.py` (byte-equal certificate regeneration from pinned source JSONs and builders);
- the universality discipline document `UNIVERSALITY_LOCK.md`;
- the sibling-notation reconciliation `SIBLING_NOTATION.md`.

On the reference machine, `verify_all.py` returns `12/12 PASS` in under 30 seconds, and `reproduce_all.py` reconstructs all twelve certificates byte-equal to the manifest modulo the timestamp field.

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